MINI PROJECT

Submitted By

Ritesh Bhaskar

05614813123

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MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

ROHINI SEC-22, NEW DELHI – 110086

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1. INTRODUCTION AND BACKGROUND

In recent years, the rapid growth of digital payment systems and online transactions has significantly increased the risk of financial fraud, especially credit card fraud. Fraudulent transactions can lead to severe financial losses for both individuals and financial institutions. Therefore, detecting such frauds accurately and efficiently has become a major concern.

Machine Learning (ML) provides powerful techniques to identify patterns and anomalies in transaction data, making it suitable for fraud detection. However, one of the major challenges in fraud detection is the **highly imbalanced nature of datasets**, where fraudulent transactions represent only a very small fraction of total transactions.

This project focuses on applying machine learning techniques to detect fraudulent transactions. It also explores different data balancing strategies such as undersampling, oversampling, and SMOTE to handle class imbalance effectively and improve model performance.

2. PROJECT OVERVIEW AND OBJECTIVES

Project Overview:

This project focuses on building a machine learning-based system for detecting fraudulent credit card transactions. The system analyzes transaction data and predicts whether a transaction is genuine or fraudulent.

To achieve better performance, the project explores multiple approaches:

- Different machine learning models
- Different data balancing techniques
- Proper evaluation metrics

The performance of each combination is compared to determine the most effective solution.

Objectives:

- To analyze and understand the characteristics of imbalanced datasets
- To perform data preprocessing and feature scaling
- To apply various data balancing techniques (undersampling, oversampling, SMOTE)
- To train multiple machine learning models (Logistic Regression, Decision Tree, Random Forest)
- To evaluate model performance using appropriate metrics such as Precision, Recall, and F1-score
- To compare results and identify the best-performing model

Problem Statement:

The goal of this project is to develop a robust machine learning model that can accurately detect fraudulent transactions while minimizing:

- False Positives (incorrect fraud alerts)
- False Negatives (missed fraud cases)

3. SYSTEM ANALYSIS AND DESIGN

Input Data:

The dataset used in this project contains credit card transaction records. Due to privacy concerns, the original features are transformed using Principal Component Analysis (PCA), resulting in features named V1 to V28.

Additional features include:

- Time: Time elapsed between transactions
- Amount: Transaction amount
- Class: Target variable (0 = Genuine, 1 = Fraud)

Output:

The system outputs a prediction indicating whether a transaction is:

- Genuine (0)
- Fraudulent (1)

System Workflow:

Data Loading → Data Cleaning → Feature Engineering →

Train-Test Split → Scaling → Handling Imbalance →

Model Training → Prediction → Evaluation → Comparison

Key Challenges:

- Highly imbalanced dataset
- Difficulty in detecting rare fraud patterns
- Risk of misleading evaluation using accuracy
- Need for robust evaluation metrics

4. TECHNOLOGY STACK AND IMPLEMENTATION

Tools and Libraries:

- Python: Main programming language
- Pandas & NumPy: Data manipulation and numerical operations
- Matplotlib & Seaborn: Data visualization
- Scikit-learn: Machine learning algorithms
- Imbalanced-learn: Handling imbalanced datasets using SMOTE

Models Used:

1. Logistic Regression

A simple linear model used as a baseline. It helps in understanding how a basic model performs on imbalanced data.

2. Decision Tree

A rule-based model that splits data into branches. It is easy to interpret but prone to overfitting.

3. Random Forest

An ensemble learning method that combines multiple decision trees. It improves accuracy and reduces overfitting.

Data Preprocessing Steps:

- Splitting dataset into training and testing sets using stratification
- Scaling numerical features (Time and Amount)
- Separating features (X) and target (y)
- Handling class imbalance using different techniques

Balancing Techniques:

Undersampling

Reduces the size of the majority class to balance the dataset.

Oversampling

Duplicates minority class samples to match the majority class.

SMOTE

Generates synthetic data points for the minority class, improving generalization.

5. DEPLOYMENT AND RESULTS

The models were trained using different balancing techniques and evaluated on the test dataset. Since the dataset is imbalanced, evaluation was based on Precision, Recall, and F1-score instead of accuracy.

Results Summary:

creditcard project fraud detection colle... Draft saved

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[2]:

	Method	Model	Precision	Recall	F1
0	Under	Logistic	0.04	0.92	0.07
1	Over	Logistic	0.06	0.92	0.12
2	SMOTE	Logistic	0.06	0.92	0.11
3	Under	Decision Tree	0.02	0.91	0.03
4	Over	Decision Tree	0.71	0.71	0.71
5	SMOTE	Decision Tree	0.34	0.78	0.47
6	Under	Random Forest	0.04	0.91	0.08
7	Over	Random Forest	0.96	0.77	0.85
8	SMOTE	Random Forest	0.85	0.83	0.84

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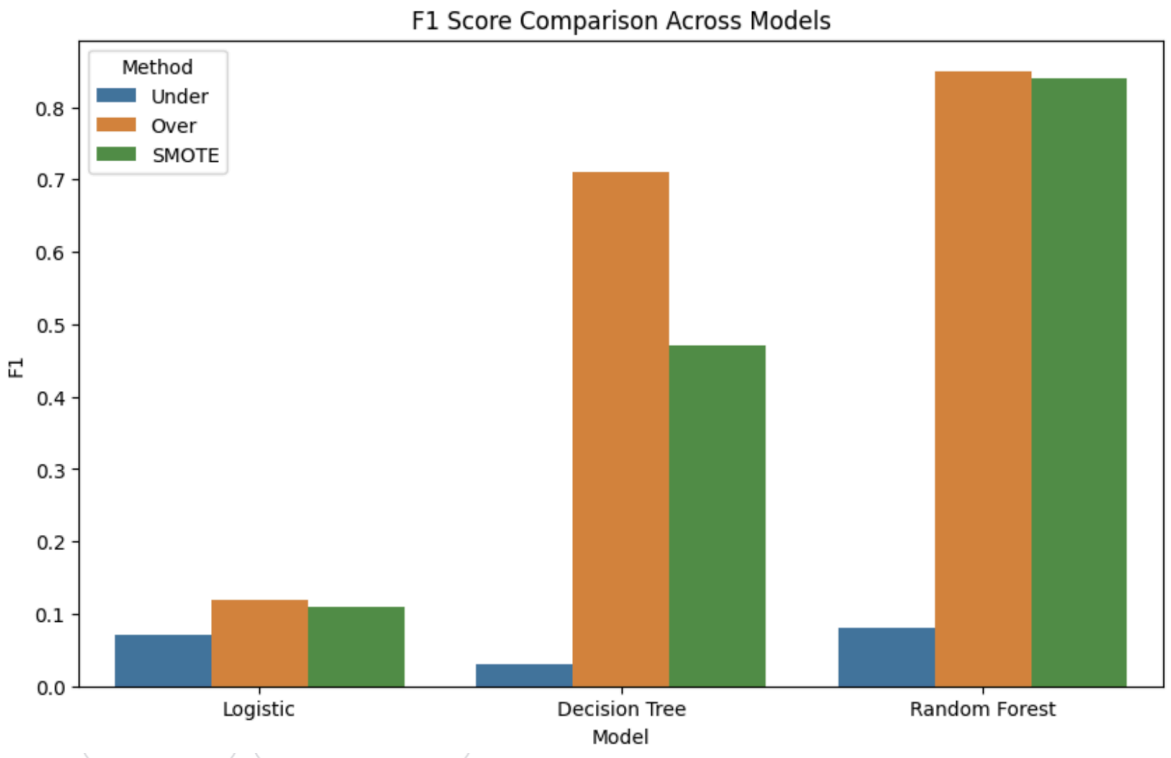
[3]:

```
pivot = df_results.pivot(index='Model', columns='Method', values='F1')
pivot
```

[3]:

Method	Over	SMOTE	Under
Model			
Decision Tree	0.71	0.47	0.03
Logistic	0.12	0.11	0.07
Random Forest	0.85	0.84	0.08




```python
best = df_results.sort_values(by='F1', ascending=False).head(1)
print("Best Model:")
print(best)
```

```
Best Model:
  Method          Model  Precision  Recall    F1
7   Over  Random Forest       0.96    0.77  0.85
```

+ Code + Markdown

Detailed Observations:

- Logistic Regression showed high recall but poor precision due to its linear nature
- Decision Tree overfitted when oversampling was used
- Random Forest performed better by reducing variance
- SMOTE helped generate better data representation

Best Model:

Random Forest with SMOTE achieved the best balance between **precision** and **recall,** making it the most suitable model.

6. CHALLENGES AND LEARNING OUTCOMES

Challenges:

- Handling highly imbalanced data
- Selecting appropriate evaluation metrics
- Avoiding overfitting in tree-based models
- Understanding trade-offs between precision and recall

Learning Outcomes:

- Importance of data preprocessing and scaling
- Understanding confusion matrix and evaluation metrics
- Handling imbalanced datasets effectively
- Comparing models scientifically
- Developing problem-solving and analytical skills

7. CONCLUSION AND FUTURE SCOPE

Conclusion:

This project highlights the importance of handling imbalanced datasets in fraud detection. It demonstrates that simply applying machine learning models is not sufficient; proper data handling techniques are equally important.

Among all models tested, Random Forest combined with SMOTE provided the best performance, offering a good balance between detecting fraud and minimizing false alarms.

Future Scope:

- Implementation of advanced models such as XGBoost
- Use of deep learning techniques
- Real-time fraud detection systems
- Integration with financial systems
- Hyperparameter tuning for optimization

8. REFERENCES

- Scikit-learn Documentation.
- Kaggle Credit Card Fraud Dataset.
- Imbalanced-learn Documentation.
- Online tutorials and academic resources.

Drive Link:

Click: **Demonstration Video**

Notebook Link: **Kaggle Notebook**